Digital Calibration Note for OPH Screen Microphysics: The Octahedral $\mathbb{Z}_2/S_3$ Patch Simulator

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Abstract

This note defines the octahedral $\mathbb{Z}_2/S_3$ simulator ladder as a digital calibration surface. The simulator gives an exact finite patch cover, explicit overlap collars, controlled defects, record writes, repair schedules, edge-sector labels, and negative controls. Its scope is fixed-cutoff interface logic and sphere-geometry charts. OPH's physical microphysics is the federated patch-carrier picture developed in the companion microphysics paper; this simulator is a calibration surface.

1 Reading Rule

The microphysics paper *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* treats the spherical screen as an observer-facing regulator chart and the implementation surface as a federation of finite observer patches.

This note has a narrower role:

1. validate finite patch, overlap, record, and repair bookkeeping;
2. test frustrated-cycle and defect behavior in an exact digital setting;
3. calibrate the edge-sector law on a finite declared interface;
4. provide reproducible benchmark names and run-manifest expectations;
5. avoid any claim that the digital sphere is the literal OPH substrate.

2 Octahedral Patch Cover

Use the octahedral cellulation of S^2 as a small digital regulator chart. Label vertices

$$X_+,\ X_-,\ Y_+,\ Y_-,\ Z_+,\ Z_-.$$

The graph has twelve edges and eight triangular faces. The patch cover is the three-patch cover

$$P_X, \qquad P_Y, \qquad P_Z,$$

where P_X contains the faces incident on X_+ or X_-, and similarly for P_Y and P_Z. The pairwise overlaps are denoted

$$B_{XY}, \qquad B_{XZ}, \qquad B_{YZ}.$$

The cover is intentionally small. It is large enough to expose pairwise overlaps, repair schedules, and a frustrated cycle, while remaining small enough for exact state and log inspection.

3 $\mathbb{Z}_2$ Pilot

The first digital pilot uses $G = \mathbb{Z}_2$. Each oriented link carries one bit or one qubit in the computational basis. The local gauge constraint is the even-parity constraint on incident links. Face holonomies are mod-two sums around triangular faces.

The pilot contains:

1. link registers on the twelve octahedral edges;
2. face-holonomy readouts;
3. one coarse cut bit for each active overlap;
4. record registers for patch-local write/compare/verify events;
5. a finite repair menu that flips declared collar links when doing so lowers the mismatch score without violating the local acceptance rule.

The $\mathbb{Z}_2$ pilot is the abelian sanity surface. It should pass clean synchronization cases, expose stale records, repair single collar flips, and refuse to hide frustrated cycles.

4 S_3 Upgrade

The first nonabelian upgrade keeps the octahedral graph and patch cover. The cut bit becomes an encoded S_3 register. Each group element is encoded in three bits with an invalid-word penalty. The vertex projector is

$$P_v^{S_3} = \frac{1}{6} \sum_{h \in S_3} U_v(h),$$

where $U_v(h)$ is the local left/right action on links incident to v.

The upgrade tests:

1. nonabelian orientation bookkeeping;
2. conjugacy-class sector labels;
3. invalid-word penalties;
4. noncommuting defect order;
5. class-sensitive edge-sector readout.

Passing the S_3 suite does not prove compact-gauge reconstruction. It only shows that the fixed-cutoff interface can carry nonabelian labels and record their repair behavior without collapsing to the abelian pilot.

5 Benchmark Families

The named calibration gates are:

1. `Z2-0-clean`: initialize the $\mathbb{Z}_2$ pilot in the clean sector and verify stable synchronization.
2. `Z2-1-stale-record`: flip one stored record after a write layer and require the next compare/repair cycle to expose it.
3. `Z2-2-collar-flip`: inject one collar-edge flip before compare and require local repair.
4. `Z2-3`: double-overlap test. Inject two collar defects on different overlaps and test schedule-insensitive repair.
5. `Z2-4`: frustrated-cycle test. Inject a cyclic obstruction and require the simulator not to fake global agreement.
6. `S3-0-clean`: initialize the S_3 build in the declared identity sector.
7. `S3-1-transposition`: inject a transposition-class defect.
8. `S3-2-threecycle`: inject a three-cycle defect.
9. `S3-3-mixed-class-cycle`: inject mixed nonabelian class data around the cycle and require class-sensitive failure reporting.

6 Run Manifest

Each digital run should write a manifest containing:

1. benchmark name and simulator commit;
2. graph, patch cover, group, encoding, and invalid-word policy;
3. repair weights, tie-break rule, and schedule;
4. per-cycle link state, face holonomies, overlap labels, record values, mismatch score, and repair move;
5. final normal form or obstruction report;
6. pass/fail decision with declared failure class.

The manifest is a digital analogue of the hardware evidence-bundle rule. It makes calibration claims auditable without confusing them with hardware evidence.

7 Negative Controls

Mandatory negative controls include:

1. stale-record control: disable record refresh and require the stale-record benchmark to fail;
2. wrong-orientation control: reverse one S_3 edge orientation and require a nonabelian failure signature;

3. no-illegal-penalty control: set the invalid-word penalty to zero and require failure in the S_3 suite;
4. fake-repair control: allow repair moves that rewrite records without touching overlap data and require the exact compare log to reject the run.

8 Boundary

This calibration note supports the fixed-cutoff theorem interface by making it executable in a small finite regulator chart. It does not identify the unique OPH UV completion, does not carry hardware evidence, does not prove the Yang–Mills repair-gap theorem, and does not promote source-only hadron predictions.